COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling

Lecture 17: Survival Analysis
Reference: ISL, Chapter 11

Introduction

- Last time: Poisson model in which Y is a count.
- Count models are fundamentally about rates
  - Rate of heads per coin toss
  - Bottles of wine sold per week
- Can also estimate rates by modeling time-to-event
- Tricky, because cannot ignore *censored* cases
  - Left-censored: Don't know when time started
  - Right-censored: Something cut observation off before event occurred
- Ignoring censored cases leads to inferential error

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- Sounds like a *regression problem*. But there is an important complication: some of the patients have survived until the end of the study. Such a patient's survival time is said to be *censored*.
- We do not want to discard this subset of surviving patients, since the fact that they survived at least five years amounts to valuable information.

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- The company might collect data on customers over some time period, in order to predict each customer's time to cancellation.
- However, presumably not all customers will have cancelled their subscription by the end of this time period; for such customers, the time to cancellation is censored.
- Survival analysis is a very well-studied topic within statistics. However, it has received relatively little attention in the machine learning community.

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- For everyone, we suppose that there is a true *failure* or *event* time T, as well as a true censoring time C.
- The survival time represents the time at which the event of interest occurs (such as death).
- By contrast, the *censoring* is the time at which censoring occurs. For example, the time at which the patient drops out of the study, or the study ends.

Survival and Censoring Times — Continued

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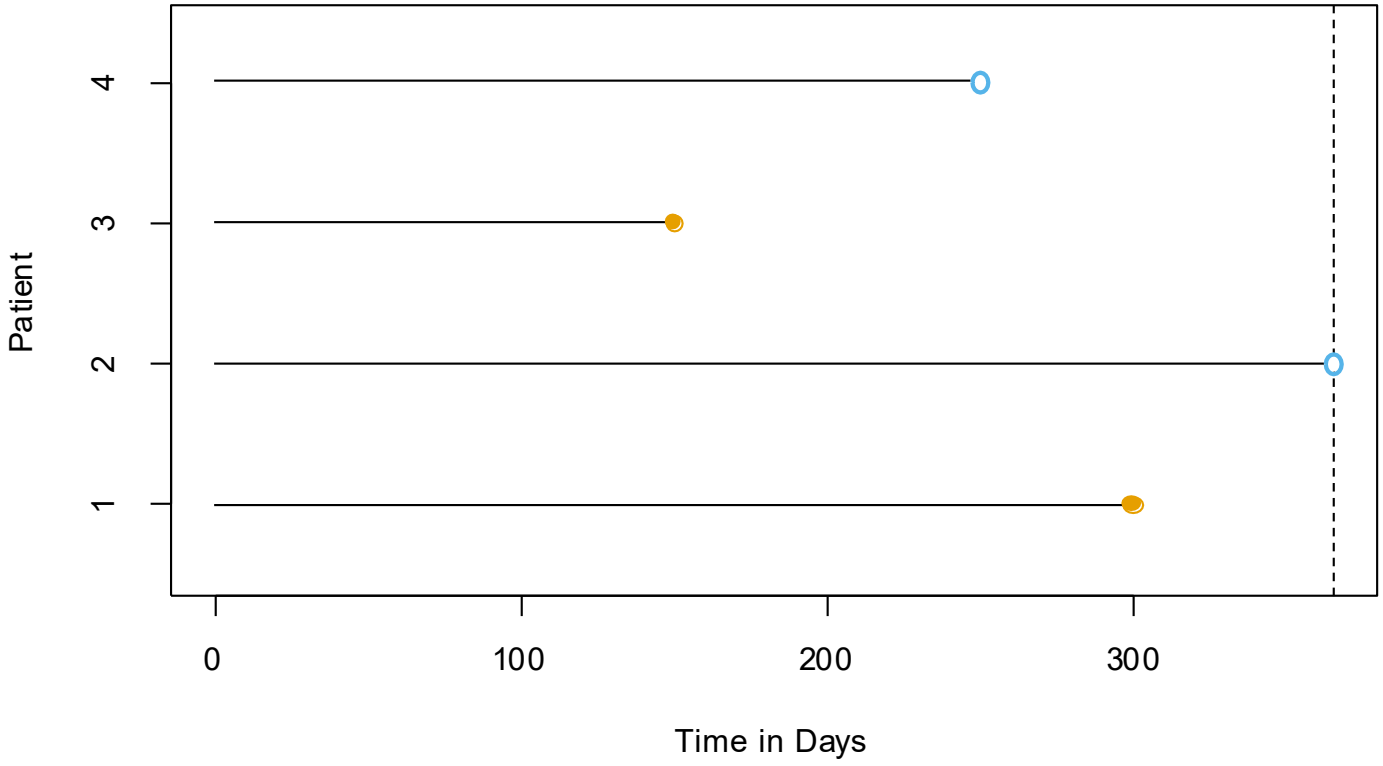
- If the event occurs before censoring (i.e. $T < C$) then we observe the true survival time T; if censoring occurs before the event ($T > C$) then we observe the censoring time. We also observe a status indicator,

$$\delta = \begin{cases} 1 & \text{if } T \leq C \\ 0 & \text{if } T > C \end{cases}$$

Finally, in our dataset we observe n pairs (Y, δ), which we denote as $(y_1, \delta_1), \ldots, (y_n, \delta_n)$.

Illustration

Here is an illustration of censored survival data. For patients 1 and 3, the event was observed. Patient 2 was alive when the study ended. Patient 4 dropped out of the study.



Therefore, $y_1 = t_1$, $y_3 = t_3$, $y_2 = c2$, $y_4 = c4$, $\delta 1 = \delta 3 = 1$, and $\delta 2 = \delta 4 = 0$.

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- An analysis that does not take into consideration the reason why the patients dropped out will likely overestimate the true average survival time.
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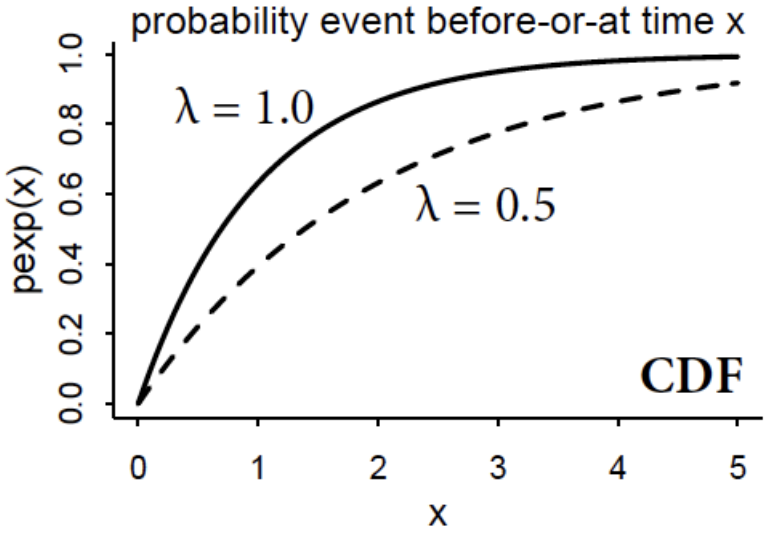
- In general, we need to assume that, conditional on the features, the event time T is *independent* of the censoring time C. The two examples above violate the assumption of independent censoring.

The Survival Curve

- Recall: *Cumulative distribution (CDF):*
- *Probability event before-or-at time x*
- Now: *Complementary cumulative distribution*
- *(CCDF): Probability not-event-yet*
- Example: Exponential distribution:

$$\Pr(D_i|\lambda_i) = 1 - \exp(-\lambda_i D_i)$$



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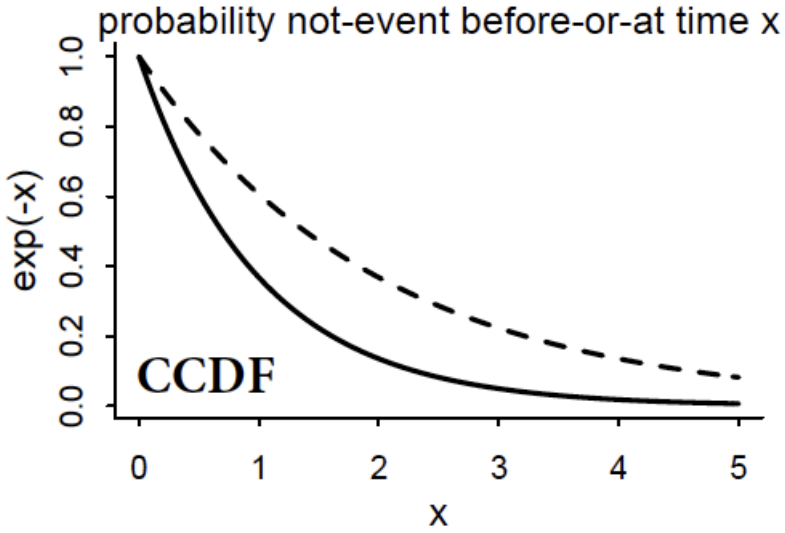

The Survival Curve

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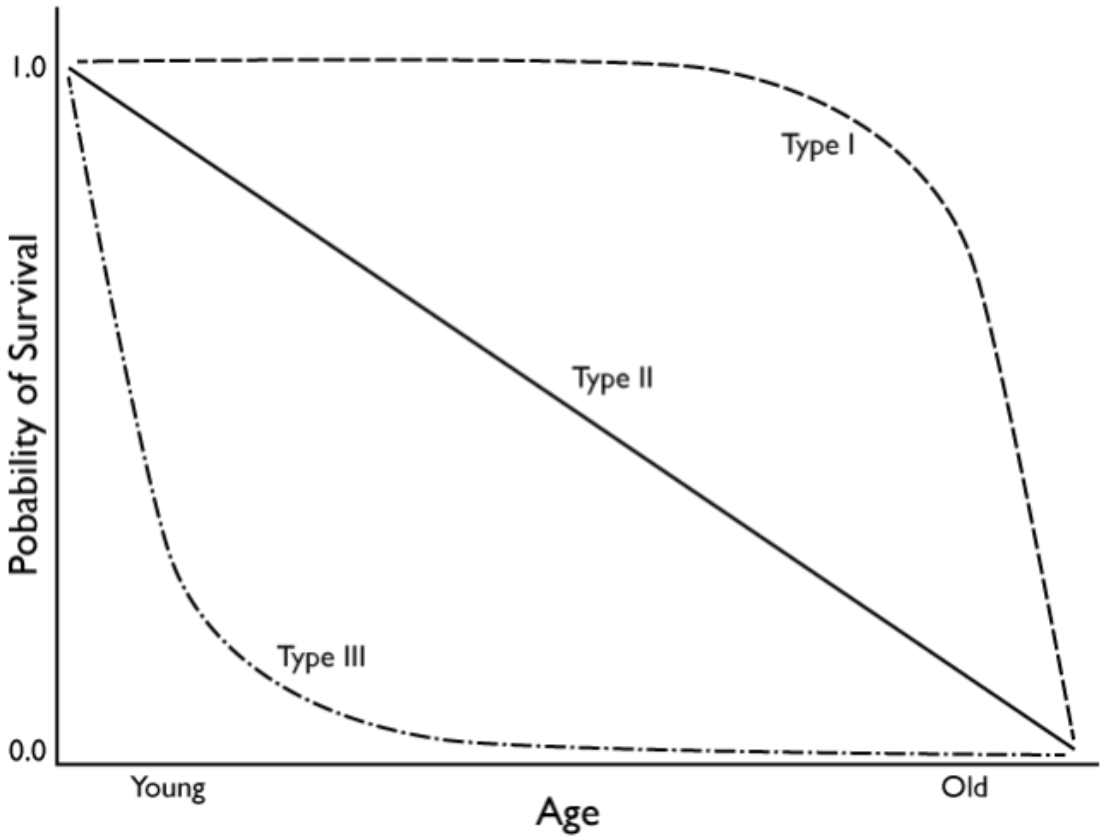
- For example, suppose that a company is interested in modeling customer churn. Let T represent the time that a customer cancels a subscription to the company's service.
- Then $S(t)$ represents the probability that a customer cancels later than time t. The larger the value of $S(t)$, the less likely that the customer will cancel before time t.

Estimating the Survival Curve



Ecologists classify species into three distinct categories based on their probability of survival. The key differences between these three patterns are related to the stage at which the probability of survival is greatest. Type I species have high probability of survival throughout life with a rapid decline late in life while Type II species are the opposite with a rapid decline early in life and a high probability of survival late in life.

Estimating the Survival Curve

- Consider the `BrainCancer` dataset, which contains the survival times for patients with primary brain tumors undergoing treatment with stereotactic radiation methods.
- The predictors are `gtv` (gross tumor volume, in cubic centimeters); `sex` (male or female); `diagnosis` (meningioma, LG glioma, HG glioma, or other); `loc` (the tumor location: either infratentorial or supratentorial); `ki` (Karnofsky index); and `stereo` (stereotactic method).

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- Only 53 of the 88 patients were still alive at the end of the study.

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Estimating the Survival Curve — Continued

- Suppose we'd like to estimate $S(20) = \Pr(T > 20)$, the probability that a patient survives for more than 20 months,
- It is tempting to simply compute the proportion of patients who are known to have survived past 20 months, that is, the proportion of patients for whom $Y > 20$.
- This turns out to be 48/88, or approximately 55%.
- However, this does not seem quite right: 17 of the 40 patients who did not survive to 20 months were censored, and this analysis implicitly assumes they died before 20 months. Hence it is probably an underestimate.

Estimating the Survival Curve — Continued

- Alternatively, to estimate $S(20)$, we could consider computing the proportion of patients for whom $Y > 20$, out of the 71 patients who were not censored by time $t = 20$; this comes out to 48/71, or approximately 68%.
- However, this is not quite right either, since it amounts to completely ignoring the patients who were censored before time $t = 20$, even though the time at which they are censored is potentially informative. For instance, a patient who was censored at time $t = 19.9$ likely would have survived past $t = 20$ had he or she not been censored.

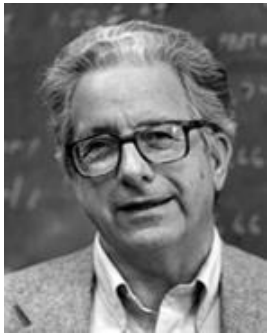
Estimate the survival curve

- Let $d_1 < d_2 < \cdots < d_K$ denote the K unique death times among the non-censored patients.
- Let q_k denote the number of patients who died at time d_k.
- For k = 1, . . . , K, we let r_k denote the number of patients alive and in the study just before d_k; these are the at-risk patients.
- Idea:

$$S(d_k) = P(T > d_k) = P(T > d_k | T > d_{k-1}) S(d_{k-1})$$
$$= P(T > d_k | T > d_{k-1}) \times \cdots$$
$$\times P(T > d_2 | T > d_1) S(d_1)$$

The Kaplan-Meier Estimate

Edward Kaplan



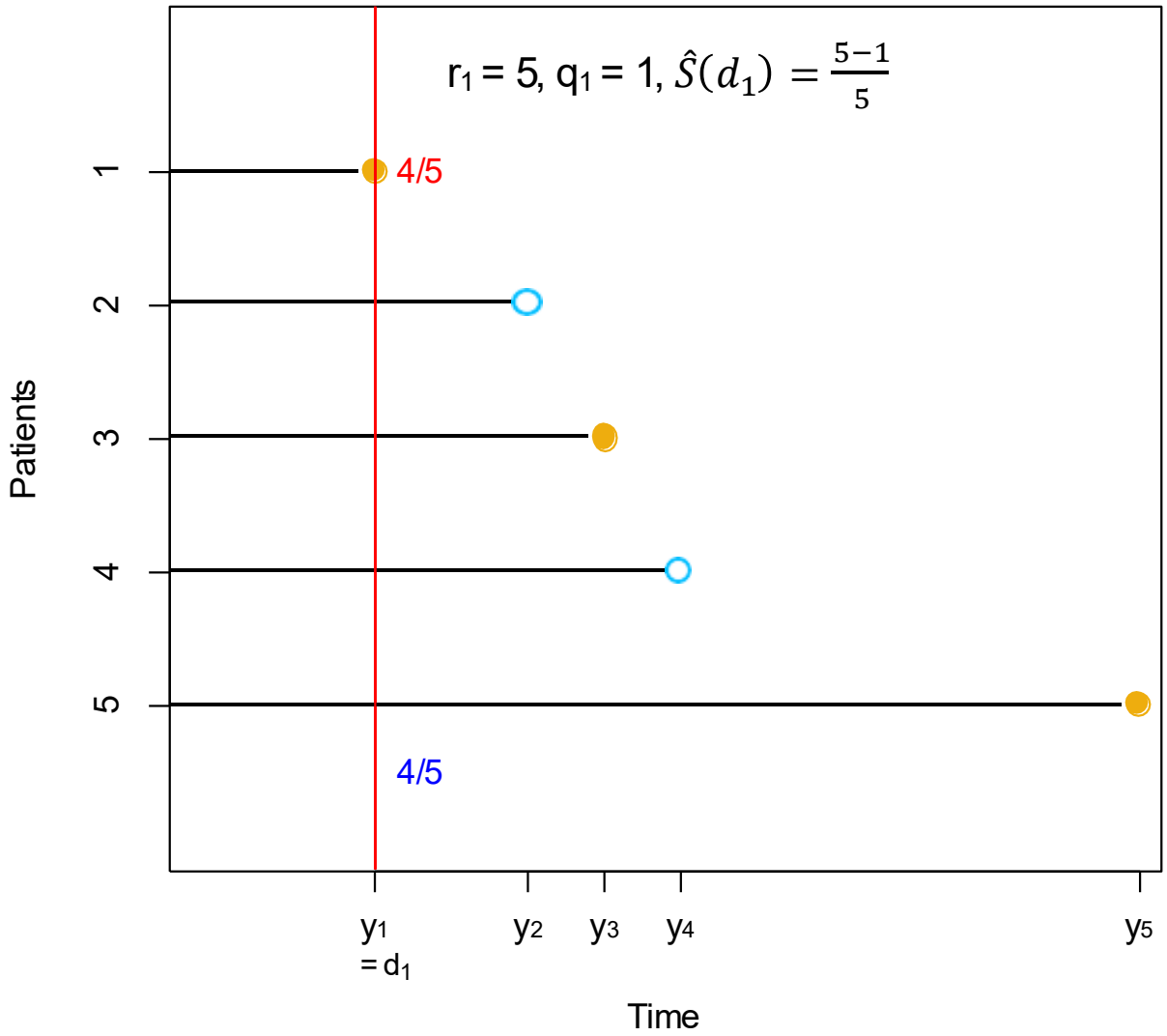
Paul Meier

$$S(d_k) = P(T > d_k | T > d_{k-1}) \times \cdots \times P(T > d_2 | T > d_1) S(d_1)$$

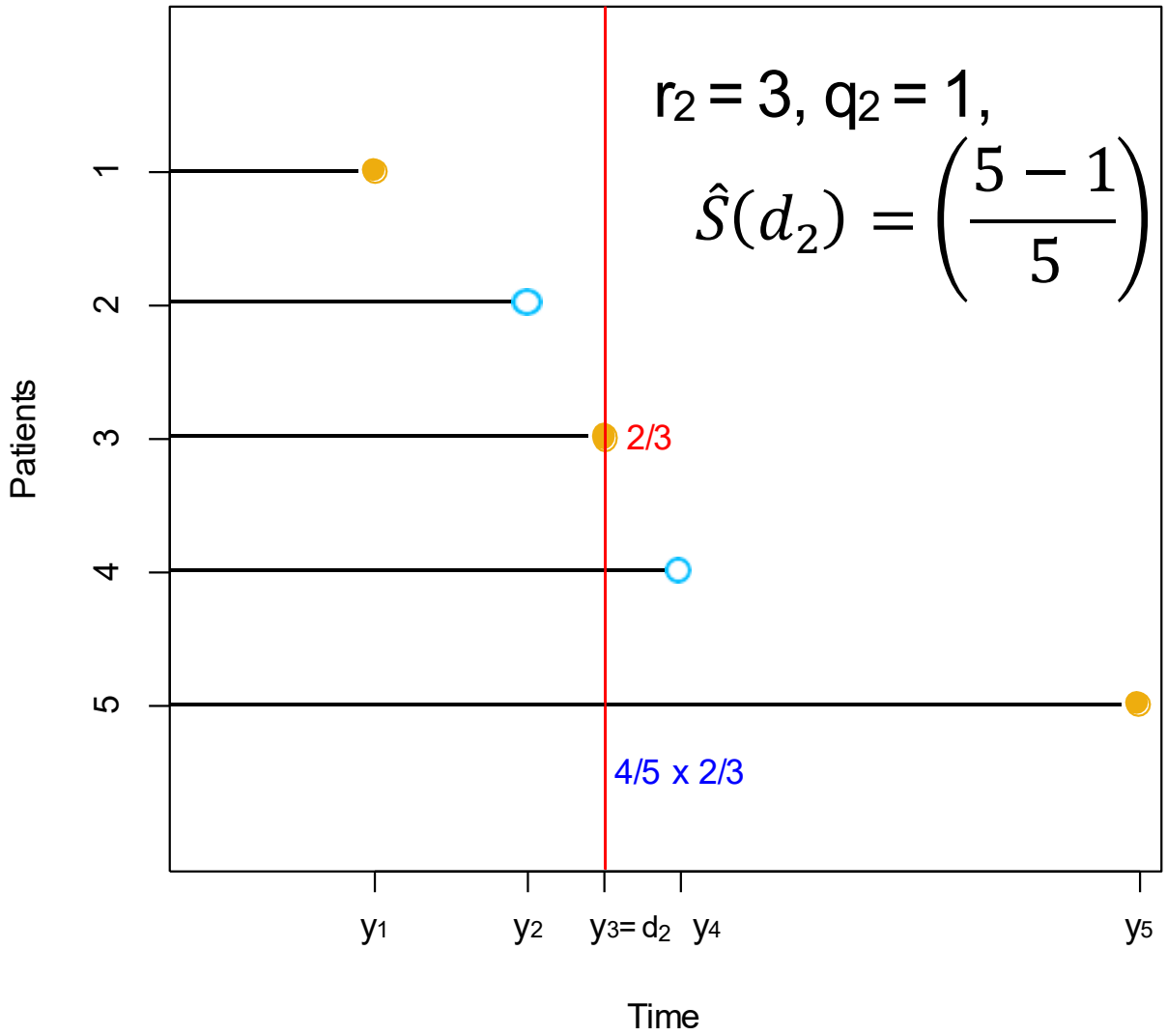
$$\hat{S}(d_k) = \prod_{j=1}^{k} \left(\frac{r_j - q_j}{r_j} \right) = \prod_{j=1}^{k} \left(1 - \frac{q_j}{r_j} \right)$$

For times t between d_k and d_{k+1}, we set $\hat{S}(t) = \hat{S}(d_k)$

First Failure



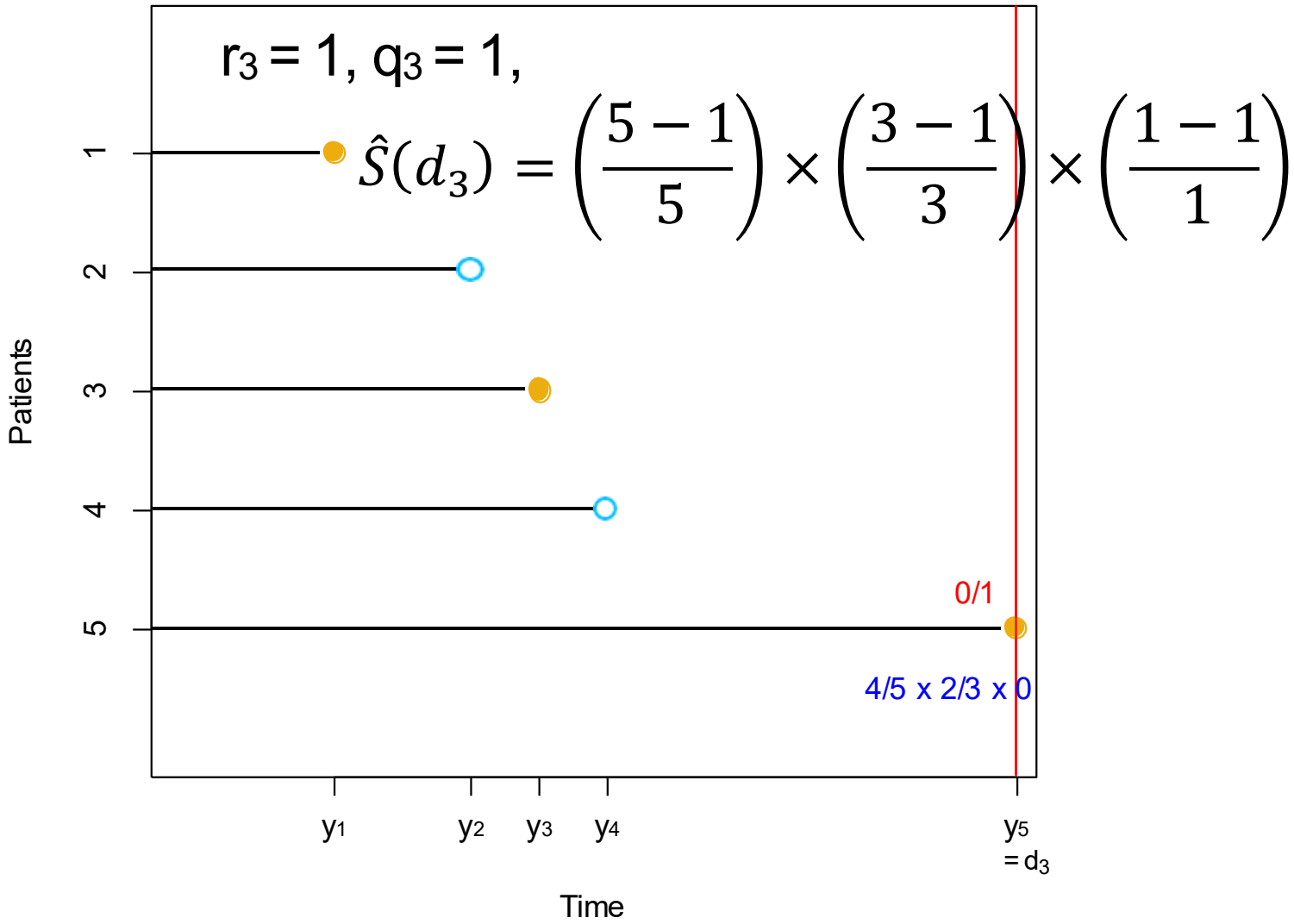
Second Failure



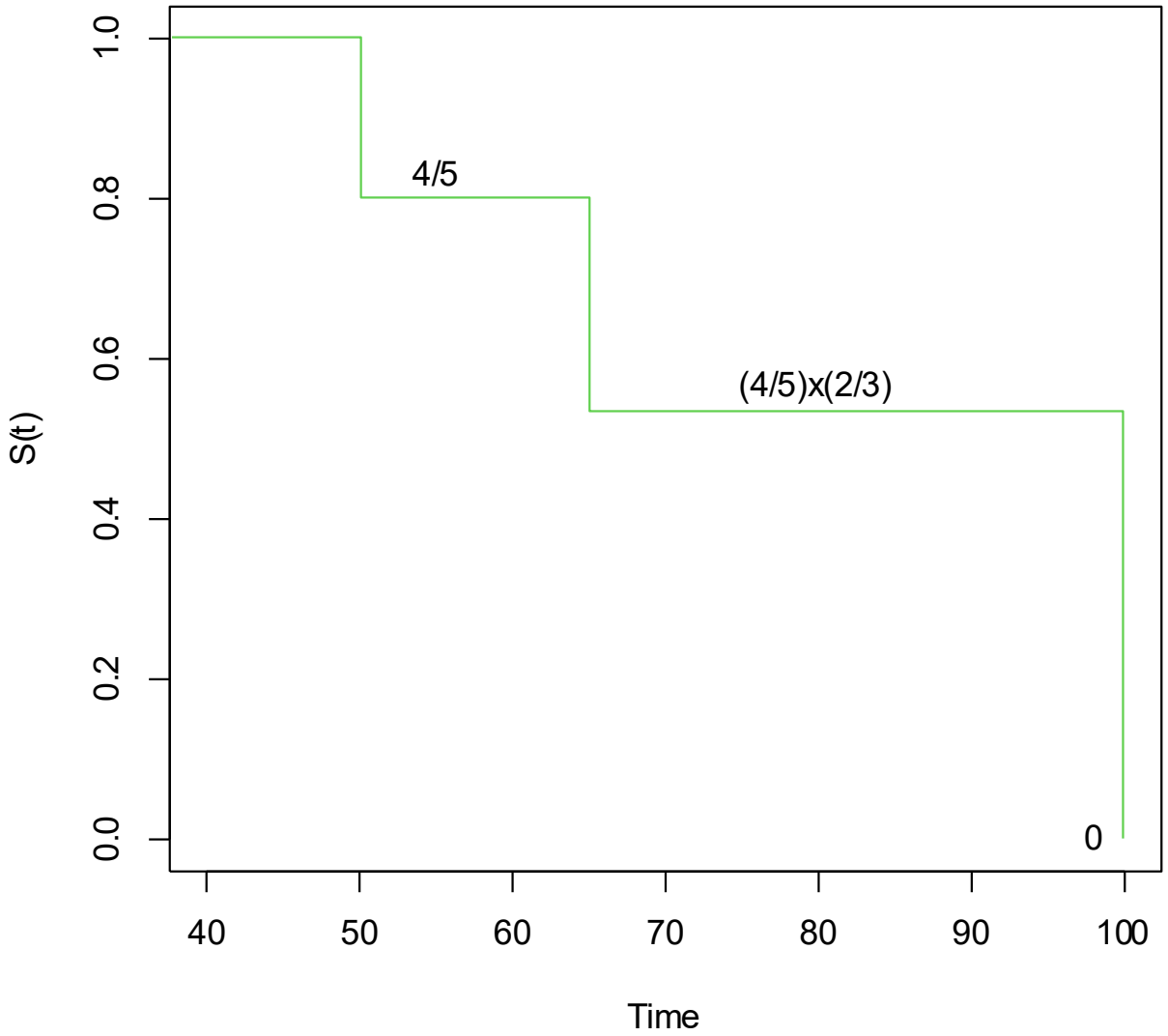
$r_2 = 3$, $q_2 = 1$,

$$\hat{S}(d_2) = \left(\frac{5-1}{5}\right) \times \left(\frac{3-1}{3}\right)$$

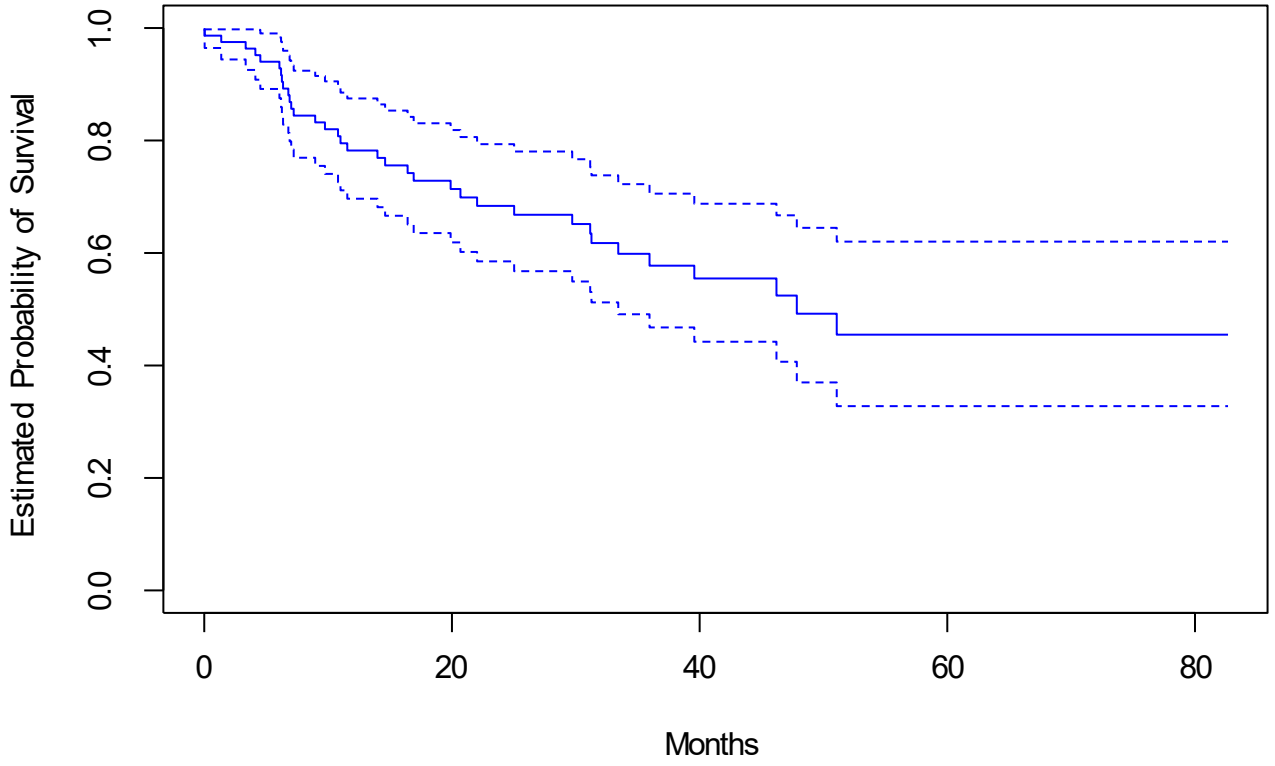
Third Failure



Resulting KM Survival Curve



Kaplan-Meier Survival Curve for the BrainCancer Data



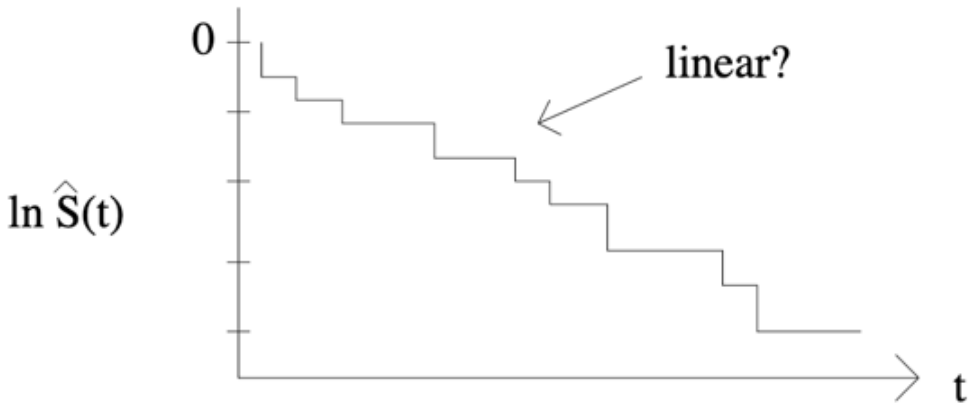
Each point in the solid step-like curve shows the estimated probability of surviving past the time indicated on the horizontal axis along with standard error bands (dashed curves).

The estimated probability of survival past 20 months is 71%, which is quite a bit higher than the naive estimate of 55% presented earlier.

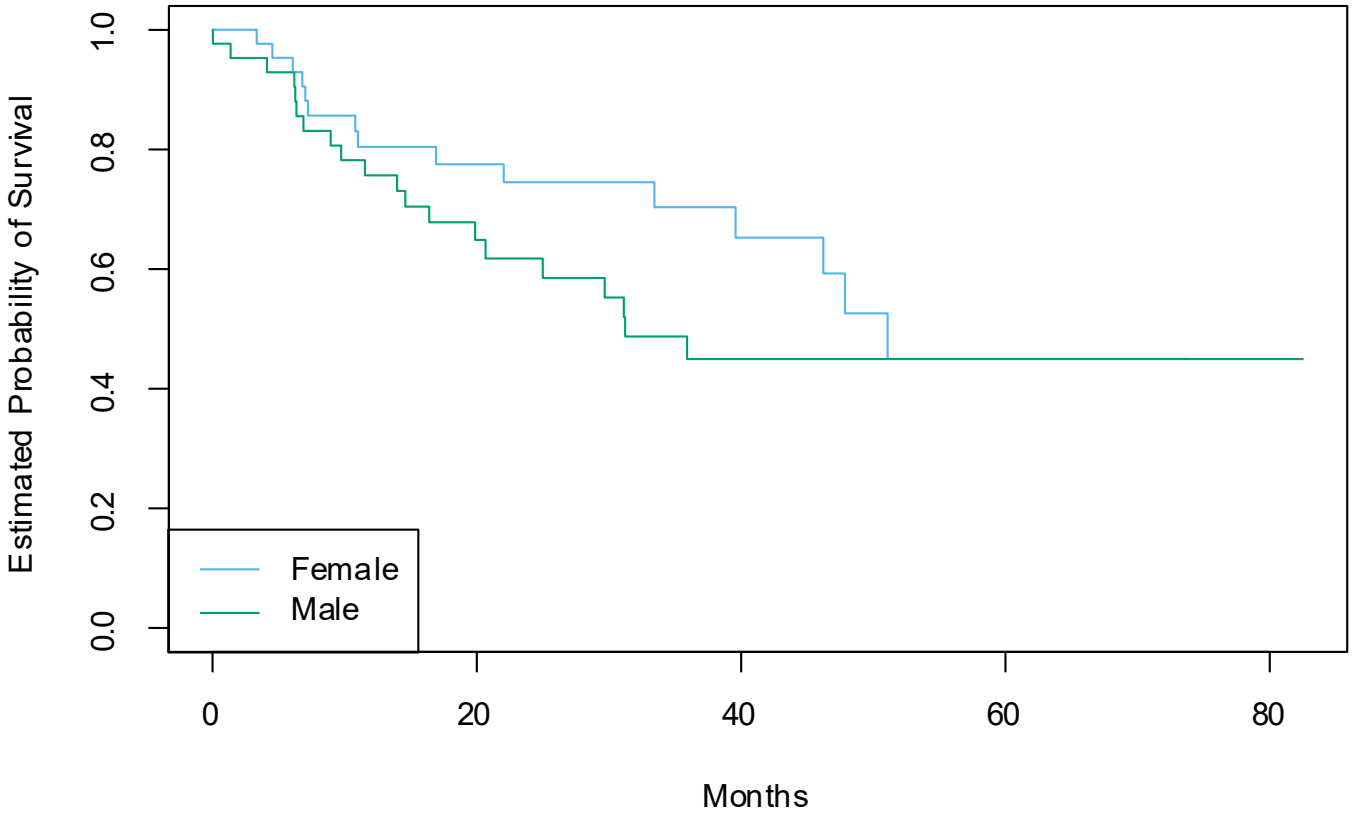
Kaplan-Meier Survival Curve

As noted earlier, the Kaplan-Meier estimator is used extensively to describe failure distributions that arise in public health and medicine. It also can be used to assess the goodness-of-fit (GOF) of a parametric assumption about F.

Example: Suppose we want to check if T ~ Exp(λ)
Then $h(t) = \lambda, S(t) = e^{-\lambda t}, \log S(t) = -\lambda t$
Hence, we can plot $\log \hat{S}(t)$ vs. t and visually check whether it appears linear; if so, data support the exponential assumption.



The Log-Rank Test



We wish to compare the survival of males to that of females. Shown are the Kaplan-Meier survival curves for the two groups.

The Log-Rank Test — Continued

Nathan Mantel **William Haenszel**

(*log rank test*)

- Females seem to fare a little better up to about 50 months, but then the two curves level off to about 50%. How can we carry out a formal test of equality of the two survival curves?
- At first glance, a two-sample t-test seems like an obvious choice: but the presence of censoring again creates a complication.

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- To overcome this challenge, we will conduct a log-rank test.

The Log-Rank Test — Continued

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- Recall that $d_1 < d_2 < \cdots < d_K$ are the unique death times among the non-censored patients, r_k is the number of patients at risk at time d_k, and q_k is the number of patients who died at time d_k.

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- We further define r_{1k} and r_{2k} to be the number of patients in groups 1 and 2, respectively, who are at risk at time d_k.
- Similarly, we define q_{1k} and q_{2k} to be the number of patients in groups 1 and 2, respectively, who died at time d_k. Note that $r_{1k} + r_{2k} = r_k$ and $q_{1k} + q_{2k} = q_k$.

Details of the Test Statistic

	Group 1	Group 2	Total
Died	q_{1k}	q_{2k}	q_k
Survived	$r_{1k} - q_{1k}$	$r_{2k} - q_{2k}$	$r_k - q_k$
Total	r_{1k}	r_{2k}	r_k

At each death time d_k, we construct a 2×2 table of counts of the form shown above.

Note that if the death times are unique (i.e. no two individuals die at the same time), then one of q_{1k} and q_{2k} equals one, and the other equals zero.

Log Rank Test: the Main Idea

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- To test $H_0 : \mathrm{E}(X) = 0$ for some random variable X, one approach is to construct a test statistic of the form

$$W = \frac{X - E(X)}{\sqrt{Var(X)}}$$

where $\mathrm{E}(X)$ and $\mathrm{Var}(X)$ are the expectation and variance, respectively, of X under H_0.

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- In order to construct the log-rank test statistic, we compute a quantity that takes exactly the form above, with

$$X = \sum_{k=1}^{K} q_{1k}$$

  where q_{1k} is given in the top left of the table above.

Details

In greater detail, under the null hypothesis of no difference in survival between the two groups, and conditioning on the row and column totals in the table, the distribution of q_{1k} is hypergeometric:

$$q_{1k} \sim HyperGeom(r_k, r_{1k}, q_k)$$

Recall: If $X \sim HyperGeom(N, K, n)$, then $P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$,

$E(X) = n\,\frac{K}{N}, Var(X) = E(X)\frac{N-K}{N}\frac{N-n}{N-1}$

Therefore, the expected value of q_{1k} is

$$E(q_{1k}) = \frac{q_k}{r_k} r_{1k}$$

Furthermore, it can be shown that the variance of q_{1k} is:

$$Var(q_{1k}) = \frac{\frac{q_k}{r_k} r_{1k}\left(1 - \frac{r_{1k}}{r_k}\right)(r_k - q_k)}{r_k - 1}$$

The Final Result

The resulting formula for the log-rank test statistic is:

$$W = \frac{\sum_{k=1}^{K}[q_{1k} - E(q_{1k})]}{\sqrt{\sum_{k=1}^{K} Var(q_{1k})}} = \frac{\sum_{k=1}^{K}\left[q_{1k} - \frac{q_k}{r_k} r_{1k}\right]}{\sqrt{\sum_{k=1}^{K} \frac{\frac{q_k}{r_k} r_{1k}\left(1 - \frac{r_{1k}}{r_k}\right)(r_k - q_k)}{r_k - 1}}}$$

When the sample size is large, the log-rank test statistic W has approximately a standard normal distribution.

This can be used to compute a p-value for the null hypothesis that there is no difference between the survival curves in the two groups.

Application to the Brain Cancer Dataset

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- The log-rank test is closely related to Cox's proportional hazards model, which we discuss next.

Regression Models with a Survival Response

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- We wish to predict the true survival time T.
  Since the observed quantity $Y = \min(T, C)$ is positive and may have a long right tail, we might be tempted to fit a linear regression of $\log(Y)$ on X. But *censoring again creates a problem*.
- To overcome this difficulty, we instead make use of a sequential construction, similar to the idea used for the Kaplan-Meier survival curve.

The Hazard Function

The *hazard function* or *hazard rate* — also known as the *force of mortality* — is formally defined as:

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t < T \leq t + \Delta t | T > t)}{\Delta t}$$

where T is the (true) survival time.

It is the death rate in the instant after time t, given survival up to that time.

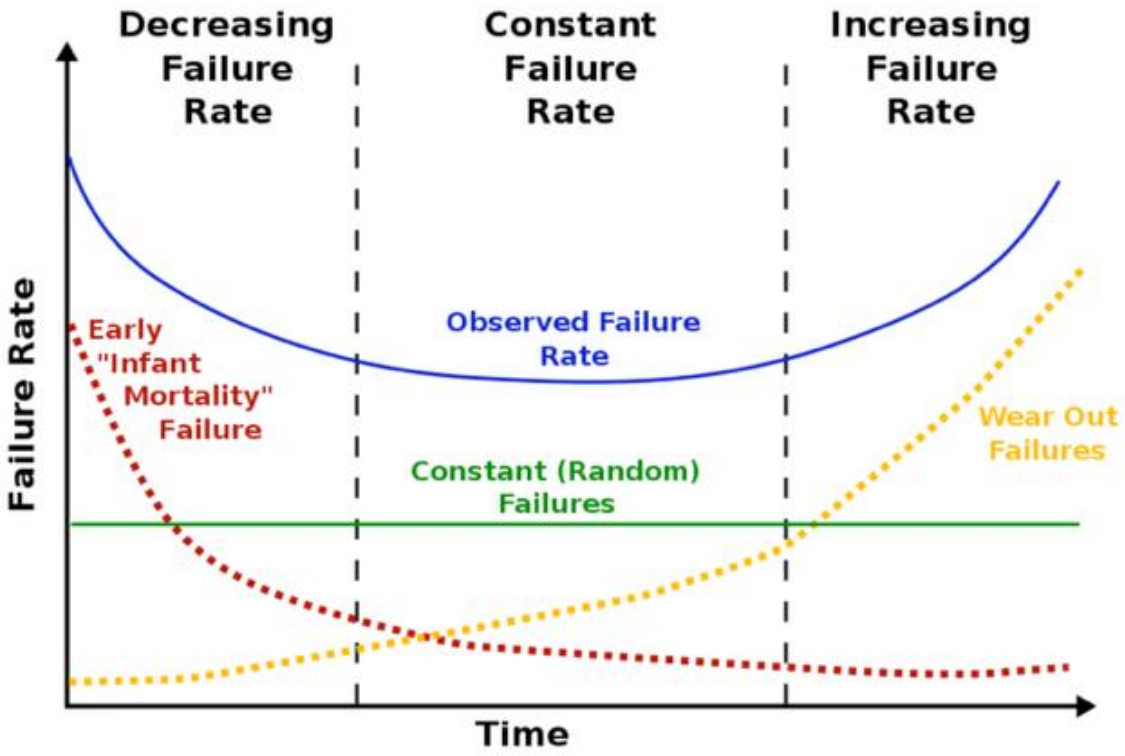
It can be shown that:

$$h(t) = \frac{f(t)}{S(t)}, \qquad S(t) = e^{-\int_0^t h(\tau) d\tau}$$

Example: If $T \sim \text{Exp}(\lambda)$, then $h(t) = \lambda$.

The hazard function is the basis for the *Proportional Hazards Model*, discussed next.

The Hazard Function

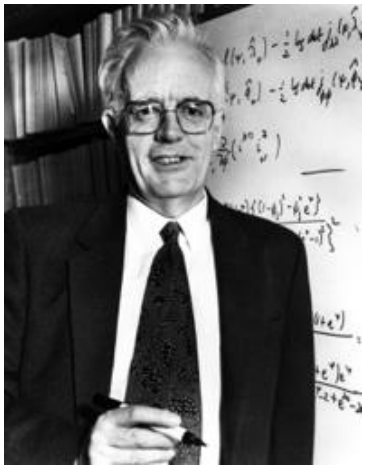


Exercise: Find the hazard function resulting from Rayleigh distribution with

$$S(t) = e^{-\theta_0 t - \frac{1}{2}\theta_1 t^2}$$

Cox's Proportional Hazards Model

- The proportional hazards assumption states that:
$$h(t|x_i) = h_0(t)e^{\sum_{j=1}^{p} x_{ij}\beta_j}$$
where $h_0(t) \geq 0$ is an unspecified function, known as the *baseline hazard*. It is the hazard function for an individual with features $x_{i1} = \cdots = x_{ip} = 0$.



David Cox

- The name *proportional hazards* arises from the fact that the hazard function for an individual with feature vector x_i is some unknown function $h_0(t)$ times the factor $e^{\sum_{j=1}^{p} x_{ij}\beta_j}$ .The quantity $e^{\sum_{j=1}^{p} x_{ij}\beta_j}$ is called the *relative risk* for the feature vector $x_i = (x_{i1}, \ldots, x_{ip})$, relative to that for the feature vector $x_i = (0, \ldots, 0)$.

Proportional Hazards Model— Continued

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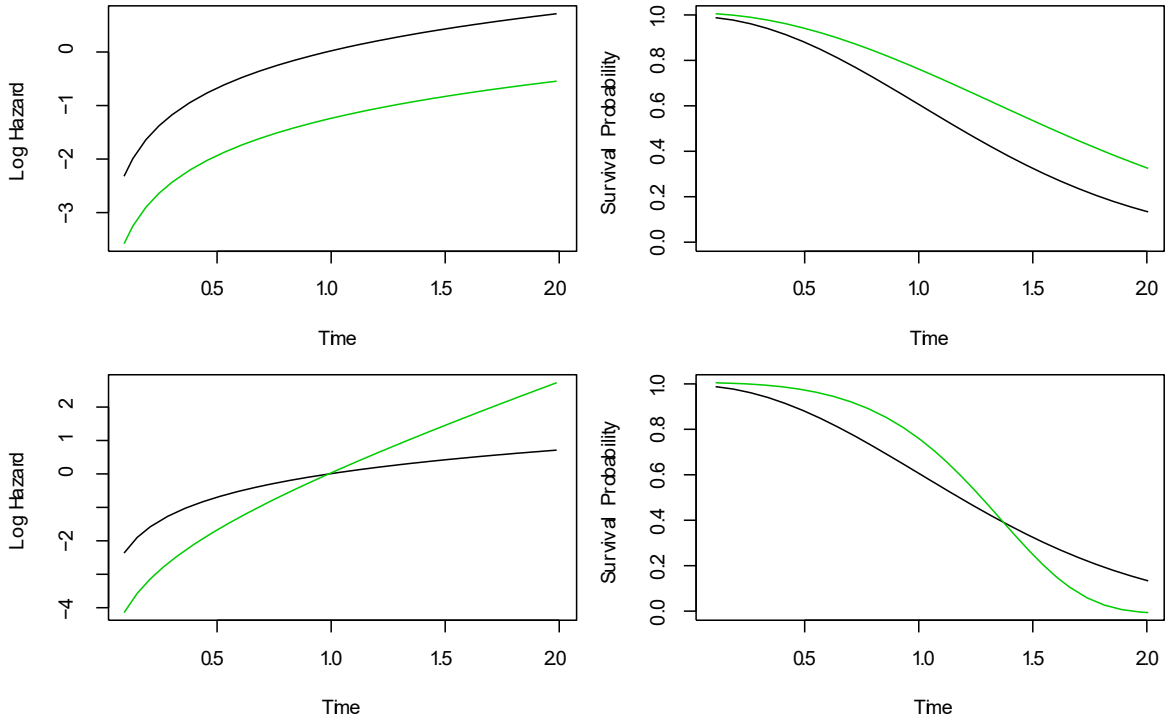
- Basically, we make *no assumptions about its functional form*. We allow the instantaneous probability of death at time t, given that one has survived at least until time t, to take any form.
- This means that the hazard function is very flexible and can model a wide range of relationships between the covariates and survival time.

Proportional Hazards Model— Continued

- What does it mean that the baseline hazard function $h_0(t)$ is unspecified?
- Basically, we make *no assumptions about its functional form*. We allow the instantaneous probability of death at time t, given that one has survived at least until time t, to take any form.
- This means that the hazard function is very flexible and can model a wide range of relationships between the covariates and survival time.
- Our only assumption is that a one-unit increase in x_{ij} corresponds to an increase in $h(t|x_i)$ by a factor of $\exp(\beta_j)$.

An Example



Here is an example with $p = 1$ and a binary covariate $x_i \in \{0, 1\}$.

Top row: the log hazard and the survival function under the model are shown (green for $x_i = 0$ and black for $x_i = 1$). Because of the proportional hazards assumption, the log hazard functions differ by a constant, and the survival functions do not cross.

Bottom row: the proportional hazards assumption does not hold.

Partial Likelihood

Partial Likelihood

- The likelihood is:

$$L = \prod_{i=1}^{n} f(y_i)^{\delta_i} S(y_i)^{1-\delta_i} = \prod_{i=1}^{n} h(y_i)^{\delta_i} S(y_i)$$

- Because the form of the baseline hazard is unknown, we cannot simply plug $h(t|x_i)$ into the likelihood and then estimate $\beta = (\beta_1, \ldots, \beta_p)^T$ by maximum likelihood.
- The magic of Cox's proportional hazards model lies in the fact that it is in fact possible to estimate β *without having to specify the form of* $h_0(t)$.
- To accomplish this, we make use of the same "sequential in time" logic that we used to derive the Kaplan-Meier survival curve and the log-rank test. Then the total hazard at failure time y_i for the at-risk observations is:

$$\sum_{i':y_{i'} \geq y_i} h_0(y_i) e^{\sum_{j=1}^{p} x_{i'j}\beta_j}$$

Partial Likelihood — Continued

Partial Likelihood — Continued

Therefore, the probability that the ith observation is the one to fail at time y_i (as opposed to one of the other observations in the risk set) is

$$\frac{h_0(y_i)e^{\sum_{j=1}^{p} x_{ij}\beta_j}}{\sum_{i':y_{i'}\geq y_i} h_0(y_i)e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

$$= \frac{e^{\sum_{j=1}^{p} x_{ij}\beta_j}}{\sum_{i':y_{i'}\geq y_i} e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

Partial Likelihood — Continued

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- Notice that the unspecified baseline hazard function $h_0(y_i)$ cancels out of the numerator and denominator!

Partial Likelihood — Continued

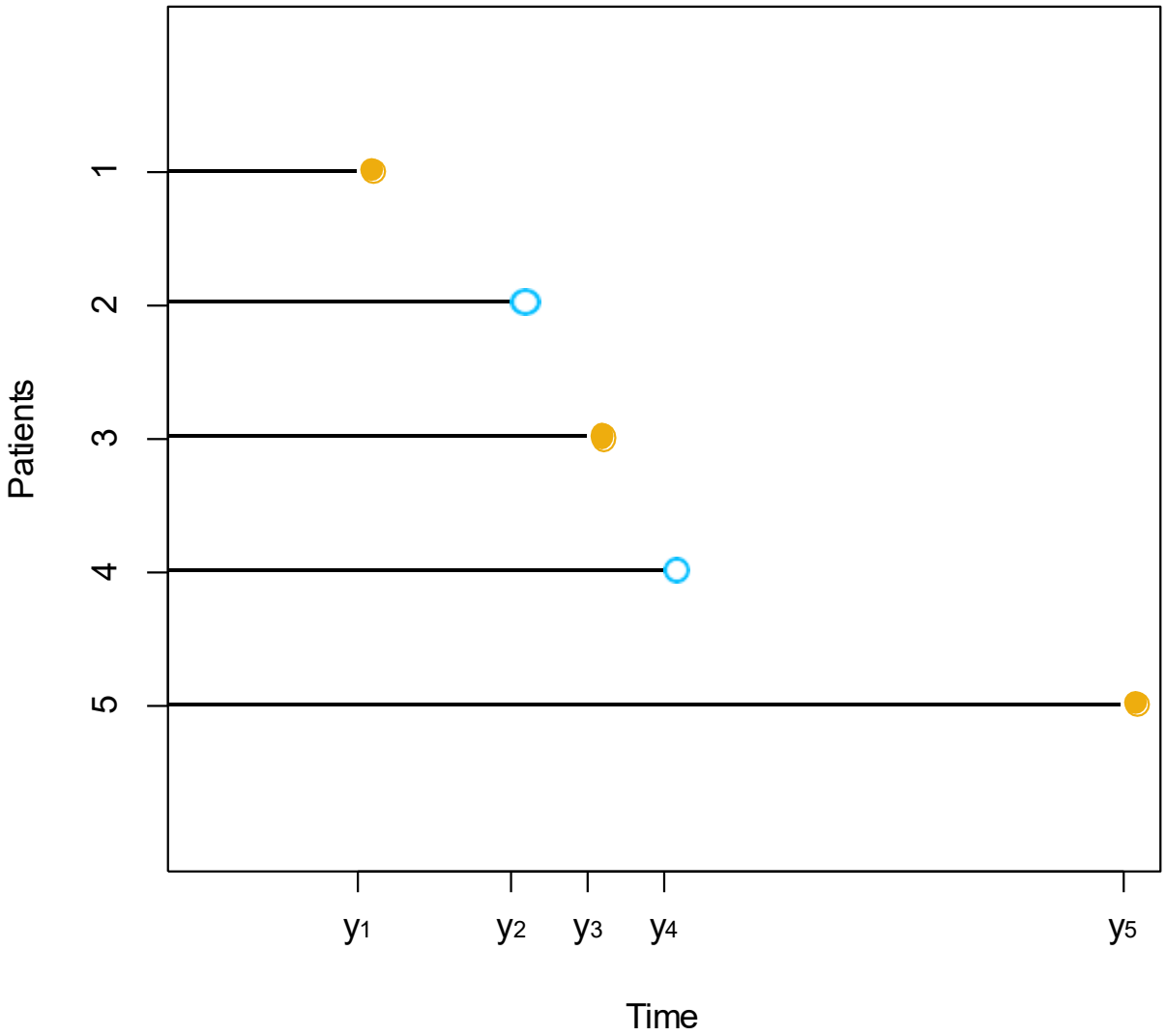
Partial Likelihood — Continued

- The partial likelihood is simply the product of these probabilities over all of the uncensored observations,

$$PL(\beta) = \prod_{i:\delta_i=1} \frac{e^{\sum_{j=1}^{p} x_{ij}\beta_j}}{\sum_{i':y_{i'}\geq y_i} e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

- Critically, the partial likelihood is valid regardless of the true value of $h_0(t)$, making the model very flexible and robust.

The Partial Likelihood: Example



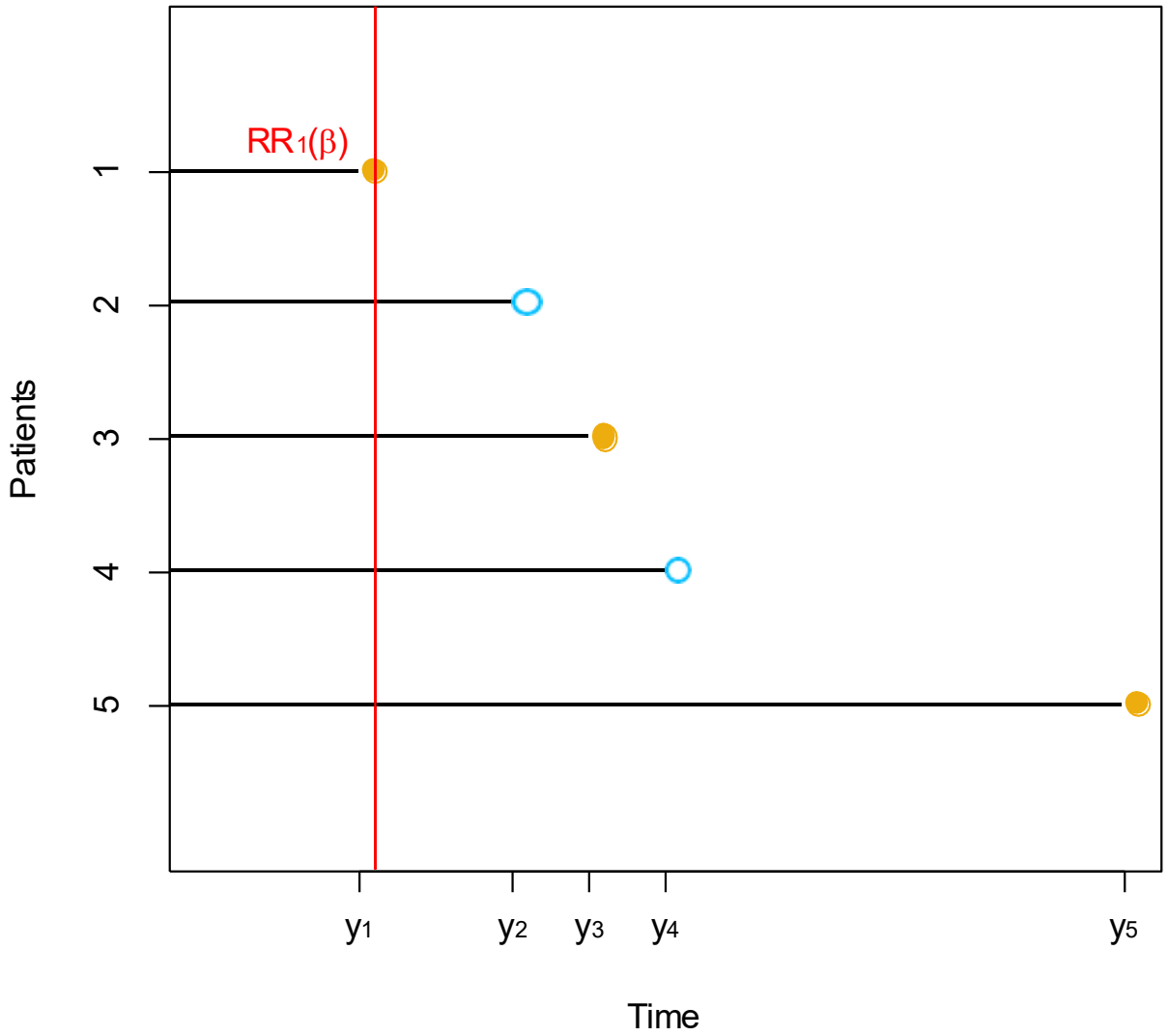
Relative Risk Functions at each Failure Time

$$RR_1(\beta) = \frac{e^{\sum_{j=1}^{p} x_{1j}\beta_j}}{\sum_{i':y_{i'}\geq y_1} e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

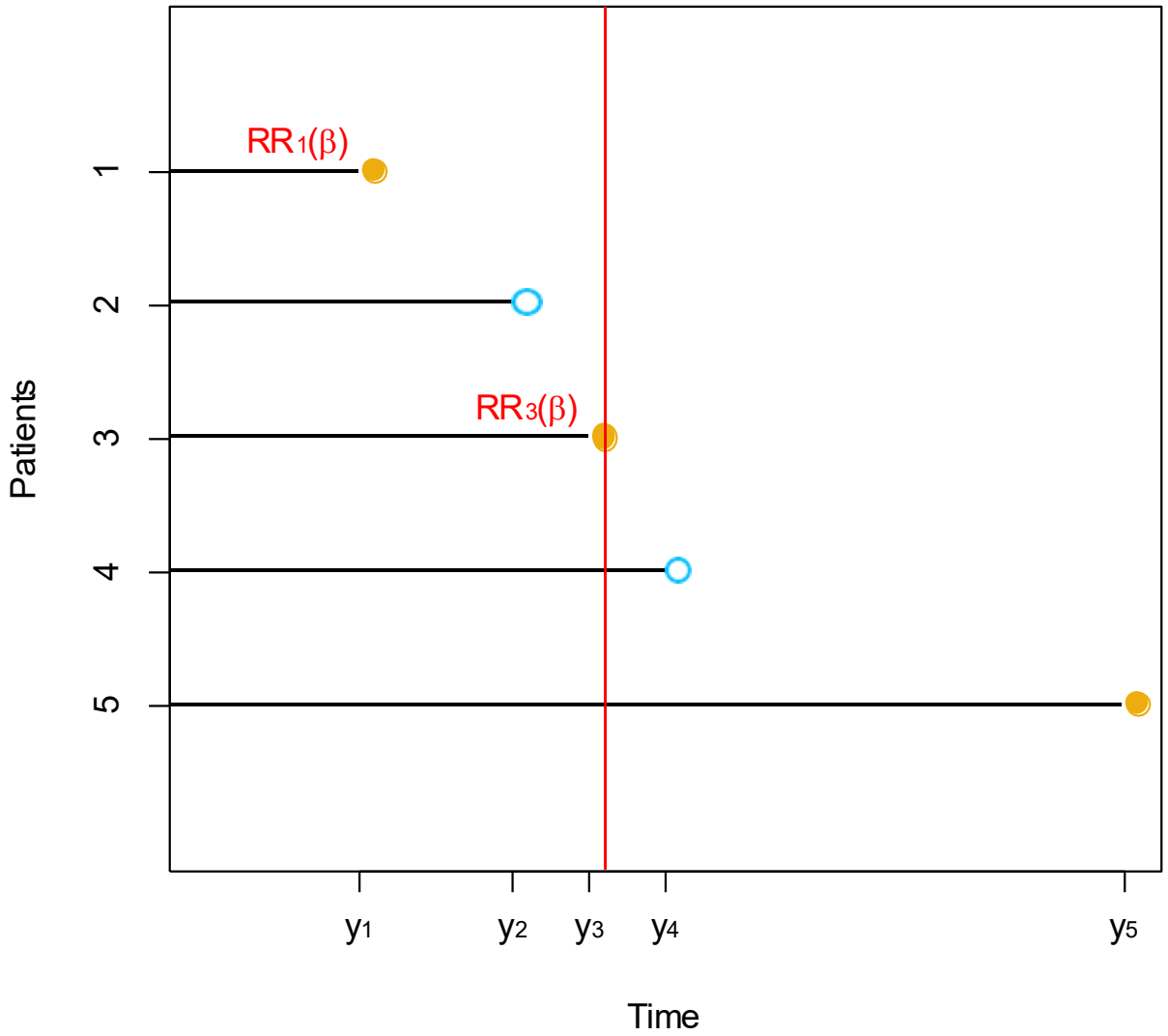
$$RR_3(\beta) = \frac{e^{\sum_{j=1}^{p} x_{3j}\beta_j}}{\sum_{i':y_{i'}\geq y_3} e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

$$RR_5(\beta) = \frac{e^{\sum_{j=1}^{p} x_{5j}\beta_j}}{\sum_{i':y_{i'}\geq y_5} e^{\sum_{j=1}^{p} x_{i'j}\beta_j}}$$

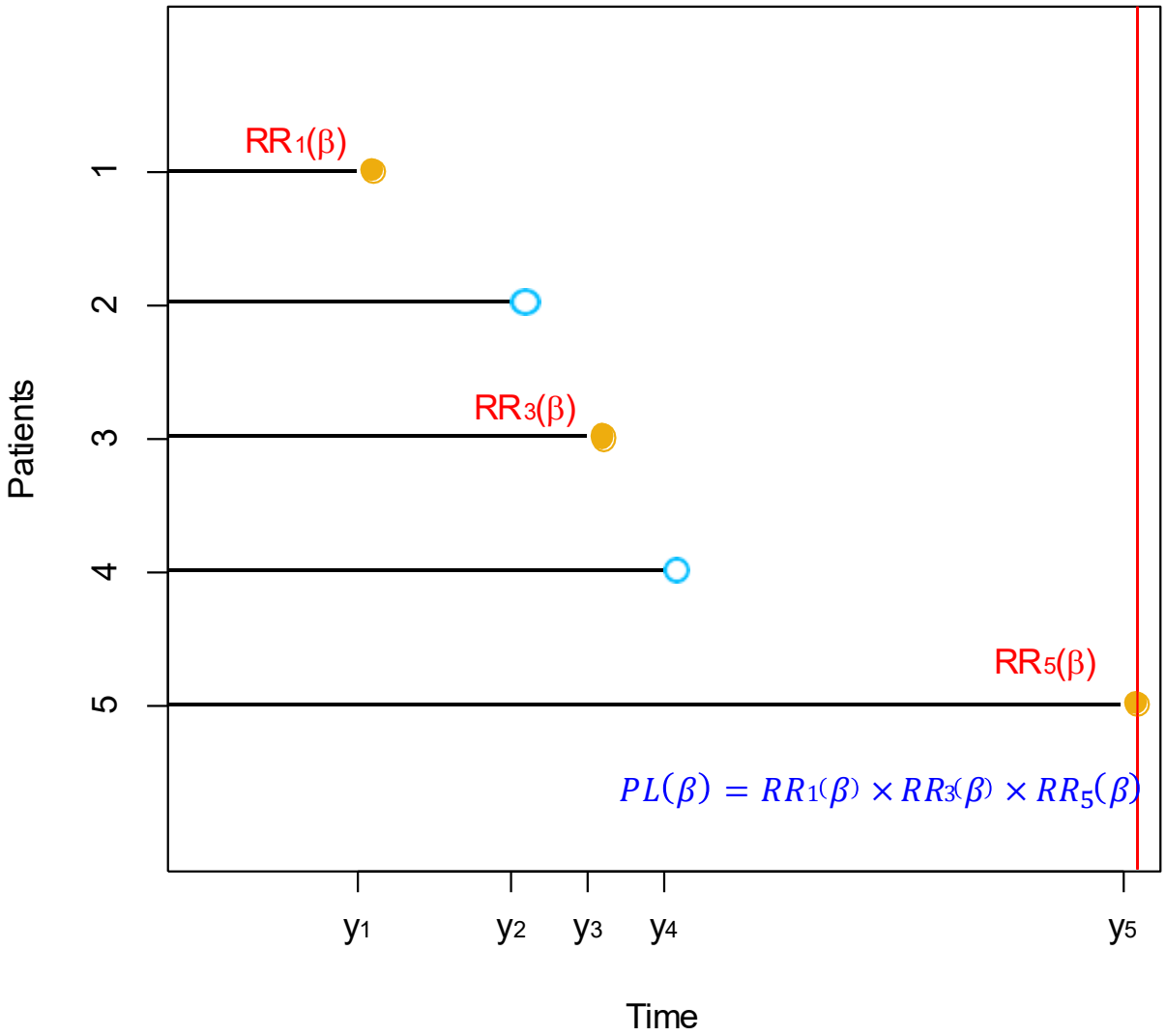
First Failure



Second Failure



Third Failure



Partial Likelihood — Computation

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- In addition to estimating β, we can also obtain other model outputs, like those in least squares regression and logistic regression.
- For example, we can obtain *p-values* corresponding to particular null hypotheses (e.g. $H_0 : \beta_j = 0$), as well as *estimated standard errors* and *confidence intervals* associated with the coefficients.

Connection with the Log-Rank Test

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- Now when taking approach #1, there are several possible ways to test H_0. One way is known as a *score test*.
- It turns out that in the case of a single binary covariate, the score test for $H_0 : \beta = 0$ in Cox's proportional hazards model is *exactly equal to the log-rank test*.

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- The *partial* likelihood gets its name because it is not exactly a likelihood. However, it is a very good approximation.
- We have focused only on estimation of the coefficients β. However, we may also wish to estimate the baseline hazard $h_0(t)$, for instance so that we can estimate the survival curve $S(t|x)$. These are implemented in the `survival` package in `R`.

Example: Brain Cancer Data

	Coefficient	Std. error	z-statistic	p-value
`sex[Male]`	0.18	0.36	0.51	0.61
`diagnosis[LG Glioma]`	0.92	0.64	1.43	0.15
`diagnosis[HG Glioma]`	2.15	0.45	4.78	0.00
`diagnosis[Other]`	0.89	0.66	1.35	0.18
`loc[Supratentorial]`	0.44	0.70	0.63	0.53
`ki`	-0.05	0.02	-3.00	<0.01
`gtv`	0.03	0.02	1.54	0.12
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- We see for example that each one-unit increase in the Karnofsky index corresponds to a multiplier of $\exp(-0.05) = 0.95$ in the instantaneous chance of dying.
- In other words, the higher the Karnofsky index, the lower the chance of dying at any given point in time. This effect is highly significant, with a p-value of 0.0027.

Exercise: Publication Data

Next, we consider the `Publication` dataset involving the time to publication of journal papers reporting the results of clinical trials funded by the National Heart, Lung, and Blood Institute.

- For 244 trials, the time in months until publication is recorded. Of the 244 trials, only 156 were published during the study period; the remaining studies were censored.

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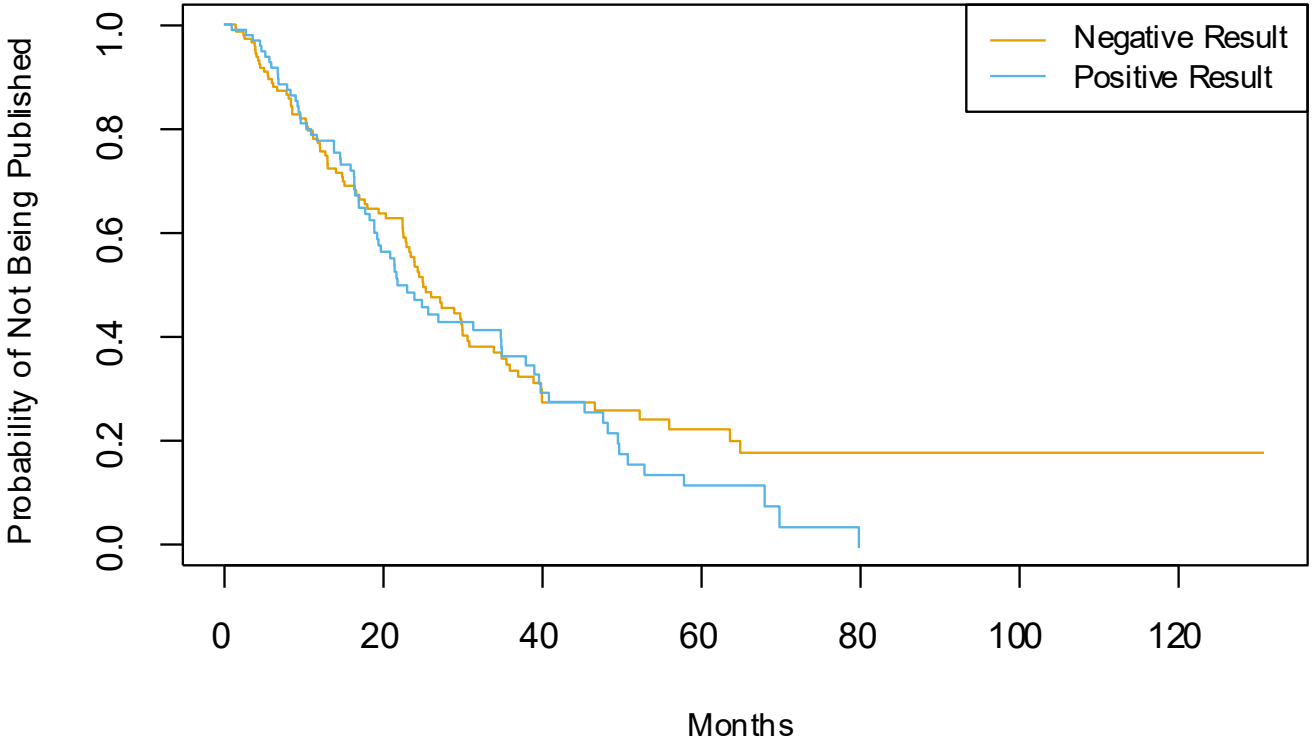
- The last covariate is particularly interesting, as a number of studies have suggested that positive trials have a higher publication rate.

Publication Data — Continued



- The figure above shows the Kaplan-Meier curves for the time until publication, stratified by whether or not the study produced a positive result.
- We see slight evidence that time until publication is lower for studies with a positive result. However, the log-rank test yields a very unimpressive p-value of 0.36.

Publication Data: Multivariate Analysis

	Coefficient	Std. error	z-statistic	p-value
`posres[Yes]`	0.55	0.18	3.02	0.00
`multi[Yes]`	0.15	0.31	0.47	0.64
`clinend[Yes]`	0.51	0.27	1.89	0.06
`mech[K01]`	1.05	1.06	1.00	0.32
many mech lines omitted				
`sampsize`	0.00	0.00	0.19	0.85
`budget`	0.00	0.00	1.67	0.09
`impact`	0.06	0.01	8.23	0.00

- The results of fitting Cox's proportional hazards model using all of the available features are shown above.
- We find that the chance of publication of a study with a positive result is $e^{0.55} = 1.74$ times higher than that of a negative result at any point in time, holding all other covariates fixed.
- The very small p-value associated with `posres` indicates that this result is highly significant.

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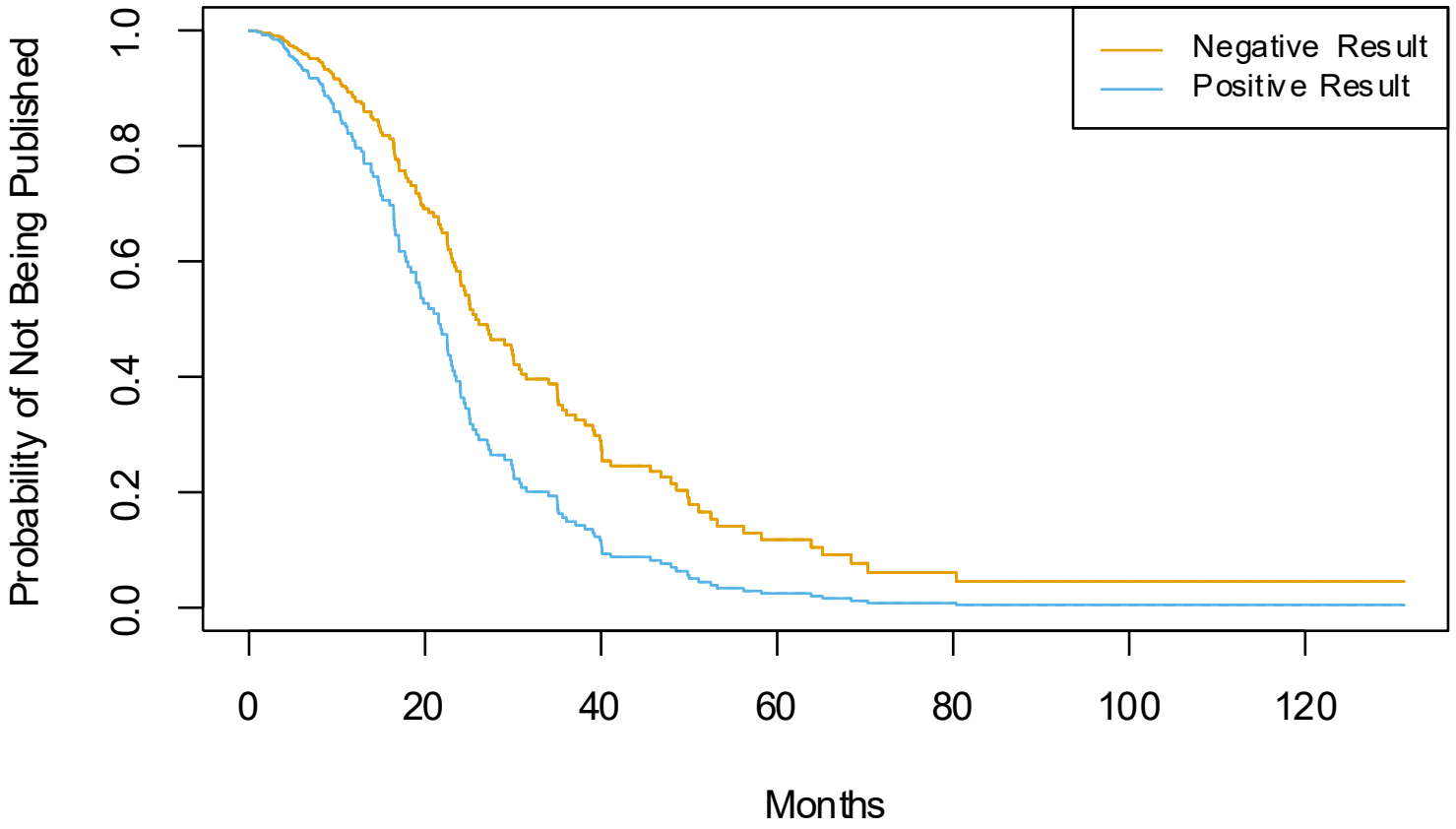
- To produce these survival curves, we estimated the underlying baseline hazard $h_0(t)$: this is implemented in the `survival` package in `R`, although the details are beyond the scope of this course.
- We also needed to select representative values for the other predictors; we used the mean value for each predictor, except for the categorical predictor `mech`, for which we used the most prevalent category (`R01`).

Adjusted Survival Curves



Adjusting for the other predictors, we now see a clear difference in the survival curves between studies with positive versus negative results. *[What has happened?]*